

Stability and Bifurcation of a Simple Food Chain in a Chemostat with Removal Rates

Mathematical Modelling Project Presentation

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Presentation Roadmap

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Core message

Distinct removal rates change the qualitative long-run behavior of the food chain.

Biological setting

A chemostat is a controlled biological reactor where continuous nutrient flow enables the study of microbial growth and predator–prey interactions.

Food chain



- ▶ S : nutrient
- ▶ x : prey
- ▶ y, z : first and second predators

Main question

- ▶ Which species survive?
- ▶ When is coexistence stable?
- ▶ When do oscillations appear?
- ▶ How do distinct removal rates reshape the outcome?

$$\begin{aligned}\frac{dS}{dt} &= 1 - S - f_1(S)x, & \frac{dy}{dt} &= y(f_2(x) - D_2) - f_3(y)z, \\ \frac{dx}{dt} &= x(f_1(S) - D_1) - f_2(x)y, & \frac{dz}{dt} &= z(f_3(y) - D_3).\end{aligned}$$

- ▶ S, x, y, z : Nutrient, prey, 1st and 2nd predator
- ▶ f_i : Monod response functions ($a_i u / (b_i + u)$)
- ▶ D_1, D_2, D_3 : Distinct removal rates
- ▶ **Key feature:** Distinct D_i removes simple conservation laws, requiring full 4D analysis.

Positivity, Boundedness, and Dissipativity

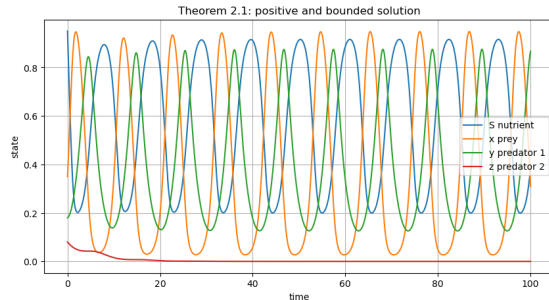
- **Positivity & Boundedness:**

S, x, y, z remain positive and bounded.
 $\min \approx 2.02 \times 10^{-9} > 0$.

- **Dissipativity:**

$1.12 \leq S + x + y + z \leq 1.62$.

- The system is dissipative rather than strictly conservative due to distinct removal rates.



S, x, y, z remain positive and bounded.

Equilibrium Points and Biological Interpretation

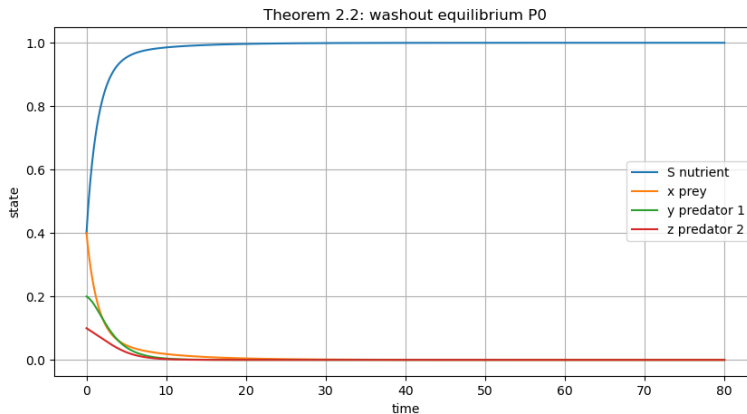
Equilibrium	Form	Biological meaning
P_0	$(1, 0, 0, 0)$	Washout of all organisms
P_1	$(S_1, x_1, 0, 0)$	Prey survives; predators extinct
P_2	$(S_2, x_2, y_2, 0)$	Prey and first predator survive
P_3	(S_3, x_3, y_3, z_3)	Full coexistence

Interpretation

Equilibria correspond to different survival or extinction patterns in the food chain.

Boundary Regimes: Washout (P_0)

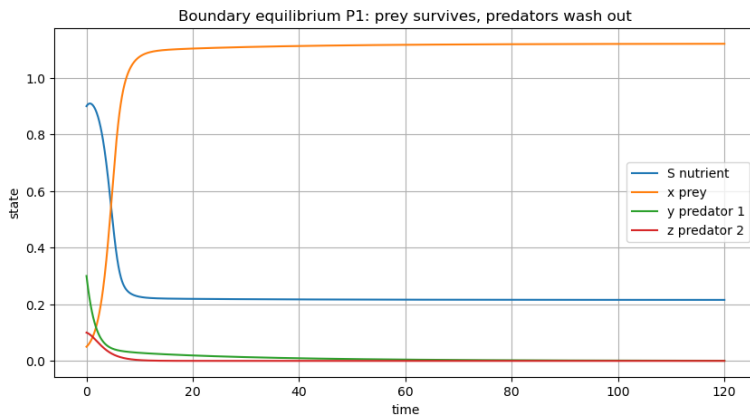
- All organisms are washed out, leaving only the nutrient at its carrying capacity.



P_0 : Washout regime

Boundary Regimes: Prey-Only (P_1)

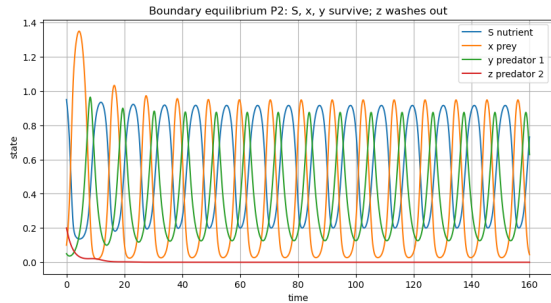
- Different parameter conditions eliminate one or more trophic levels.



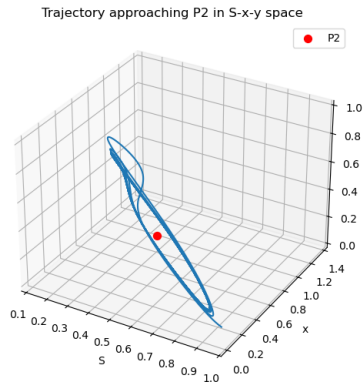
P_1 : prey-only regime

Boundary Regimes: Top Predator Washed Out (P_2)

- ▶ Boundary equilibria are observable ecological outcomes in the dynamical system.



P_2 : Time-series



P_2 : Phase portrait

Local Stability Analysis

Eq.	Main condition	Value	Interpretation
P_0	Prey invasion rate at nutrient-only state	$\max \operatorname{Re} \lambda_{\text{eig}} = -0.1333$	Washout
P_1	First-predator invasion rate at prey-only state	$\max \operatorname{Re} \lambda_{\text{eig}} = -0.0351$	Prey-only survival
P_2	Top-predator invasion condition	$f_3(y_2) - D_3 = -0.1178$	z is unable to persist near the boundary state

Careful interpretation

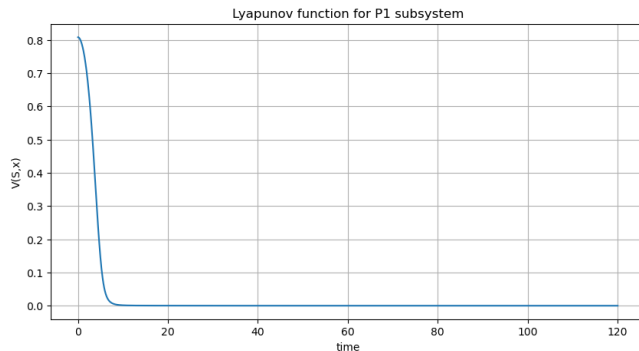
The negative top-predator invasion value for P_2 means z is eliminated near the boundary state.

Global Stability and Lyapunov Functions (1/2)

- ▶ Lyapunov profiles show an overall decrease toward small values, supporting global stability.

Prey-only subsystem

$$V(S, x) = \frac{1}{2}(S - S_1)^2 + \frac{1}{2}(x - x_1)^2$$

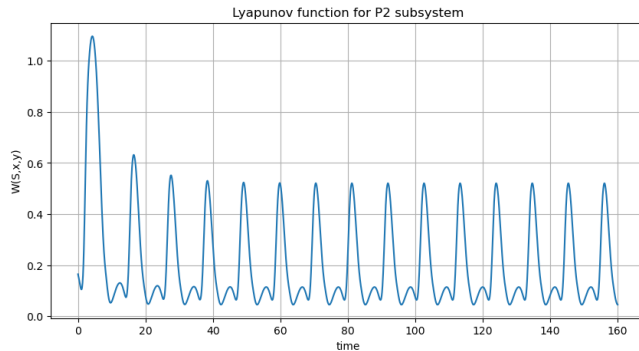


Global Stability and Lyapunov Functions (2/2)

- Extends to the boundary subsystem without a simple conservation law.

Boundary subsystem (P_2)

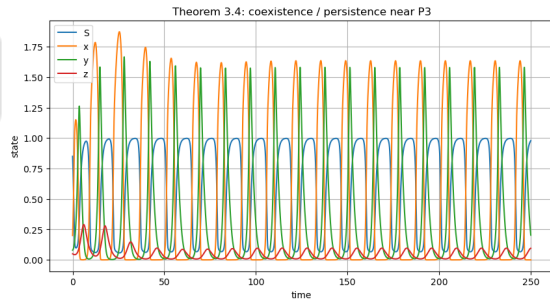
$$W = S - S_2 - S_2 \ln \frac{S}{S_2} + \frac{1}{2}(x - x_2)^2 + \frac{1}{2}(y - y_2)^2$$



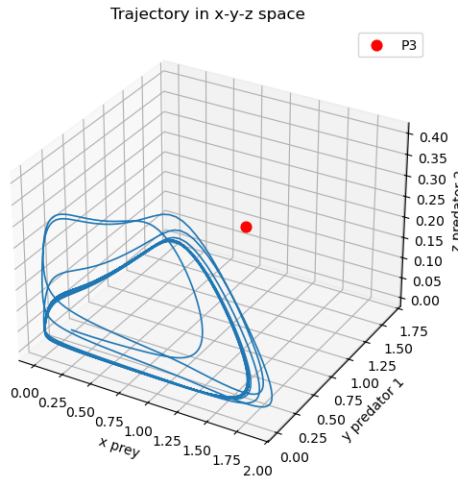
Coexistence equilibrium

$$P_3 = (0.0669, 1.6291, 0.1667, 0.3963).$$

- ▶ All four components remain active.
- ▶ Persistence ensures no species becomes extinct.

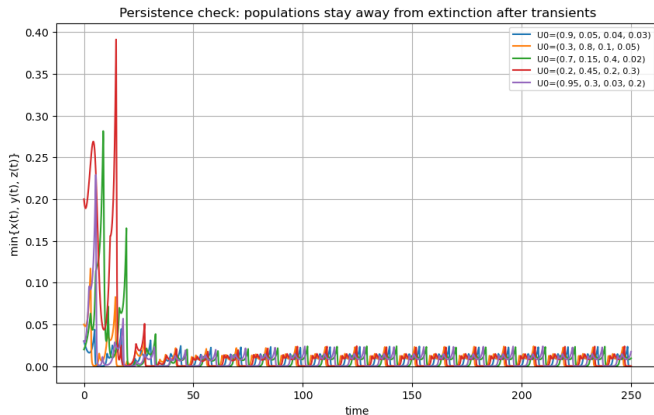


Coexistence: Phase Portrait



Observation

Coexistence-type behavior is not an artifact of a single starting point.

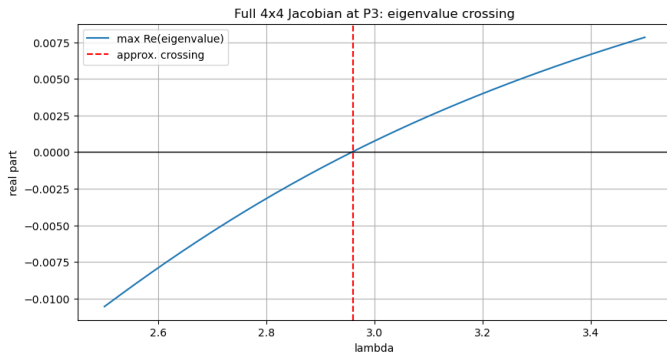


Multiple positive initial conditions

Hopf Bifurcation Analysis

Theoretical Idea

A Hopf bifurcation (complex conjugate eigenvalues crossing the imaginary axis) marks a transition to sustained oscillations, showing coexistence can be oscillatory.

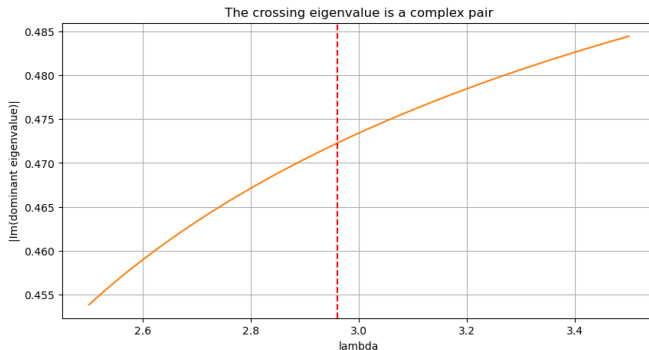


Real parts near P_2 crossing

Hopf Bifurcation Analysis (cont.)

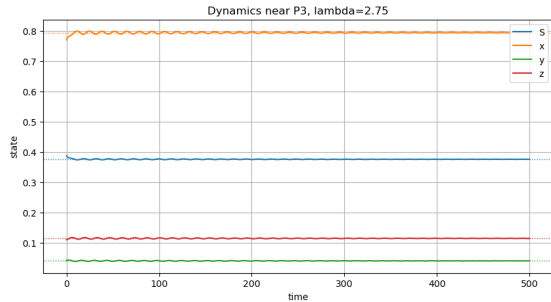
Theoretical Idea

A Hopf bifurcation marks a transition to sustained oscillations.

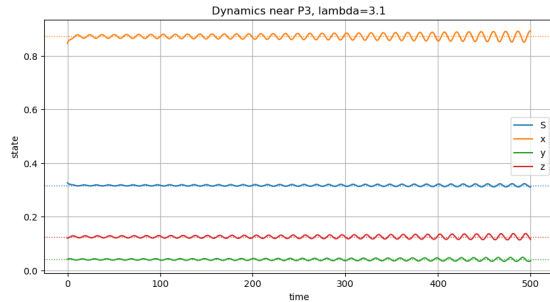


Real parts near P_3 crossing

Hopf Bifurcation Around P_2 : Time-Series



$\lambda = 8.5$: oscillatory side

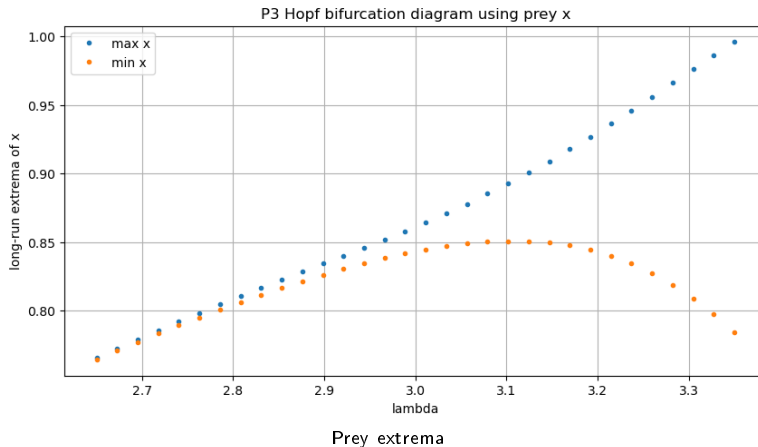


$\lambda = 9.8$: stable side

Hopf Bifurcation Around P_2 : Bifurcation Diagram

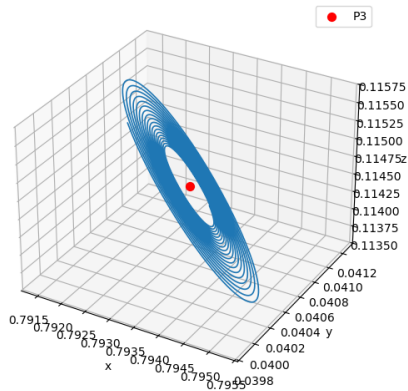
Ecological interpretation

Oscillatory boundary dynamics vs. convergence toward the boundary equilibrium.



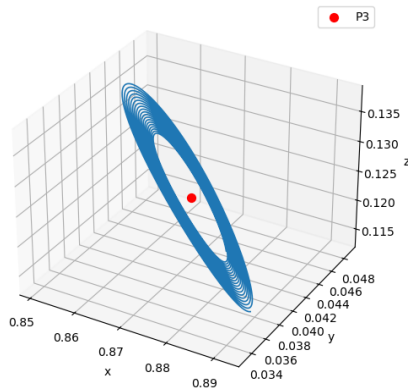
Hopf Bifurcation Around P_3 : Time-Series

Long-run phase portrait near P_3 , $\lambda=2.75$



$\lambda = 2.75$: stable side

Long-run phase portrait near P_3 , $\lambda=3.1$

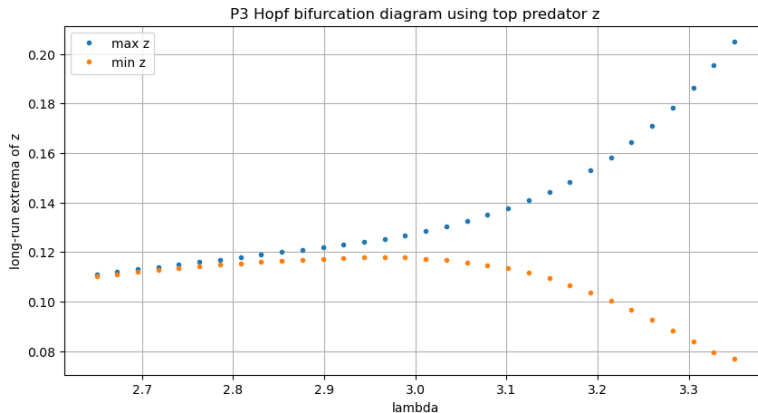


$\lambda = 3.10$: oscillatory side

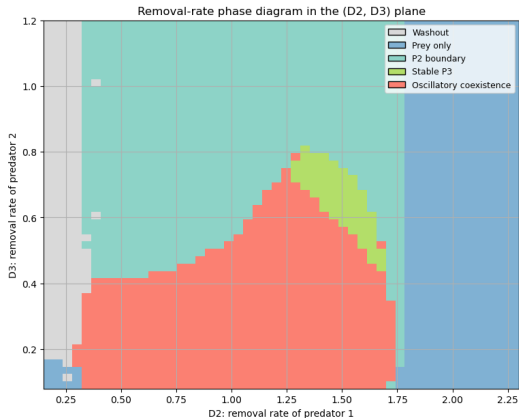
Hopf Bifurcation Around P_3 : Bifurcation Diagram

Ecological interpretation

Transition from attraction toward the coexistence equilibrium to oscillatory coexistence in the positive state space.



Effect of Distinct Removal Rates: Phase Diagram



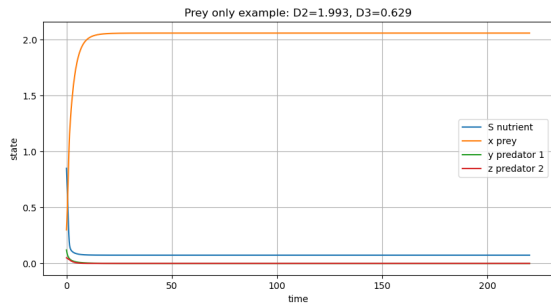
Parameter sweep

$$D_2 \in [0.15, 2.30], \quad D_3 \in [0.08, 1.20].$$

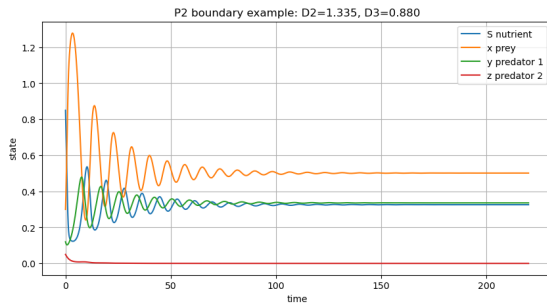
- ▶ Grid size: 50×50 .
- ▶ Five long-run regimes are classified.
- ▶ D_2 , D_3 reshape the qualitative ecological outcome.

Representative Regimes: Extinction & Boundary

- Large predator removal rates eliminate higher trophic levels.



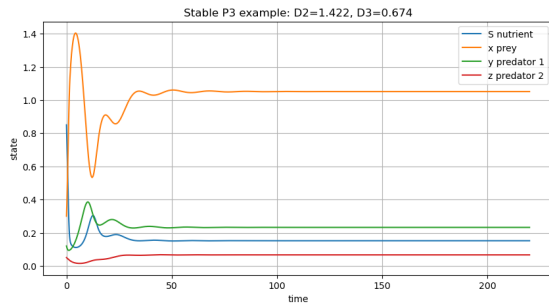
Prey-only



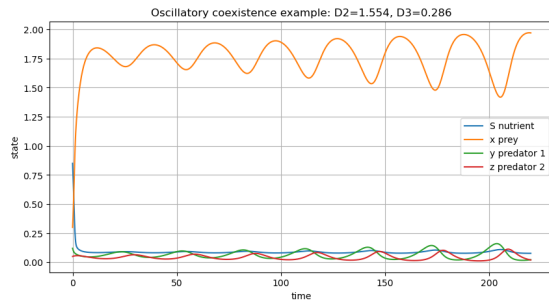
P_2 boundary

Representative Regimes: Coexistence

- Some regions support full coexistence; others produce sustained oscillations.



Stable P_3



Oscillatory coexistence

Main conclusions

- ▶ Positivity, boundedness, and dissipativity are numerically supported.
- ▶ Boundary regimes match expected ecological regimes and invasion conditions.
- ▶ Hopf bifurcation provides visual evidence for steady vs. oscillatory dynamics in P_2 and P_3 .
- ▶ Distinct removal rates directly reshape the qualitative ecological outcome.

Limitations

- ▶ No empirical calibration data.
- ▶ Deterministic model only.
- ▶ No delay, stochastic noise, spatial structure, or external perturbations.

Distinct removal rates are not only a technical modification; they reshape the outcome.